

GRAYLING AAF, MI, USA

WMO: 722093

Lat: 44.680N

Lon: 84.729W

Elev: 350

StdP: 97.19

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.9	-19.8	-27.3	0.3	-23.7	-22.8	0.5	-19.0	10.6	-4.8	9.3	-4.3	0.2	320	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	31.2	21.1	29.4	20.1	27.5	18.8	22.8	27.7	21.8	26.3	20.9	24.8	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.7	24.2	20.5	15.8	23.7	19.7	15.1	22.8	68.9	27.7	65.5	26.0	62.0	25.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.6	6.8	-30.3	33.5	4.2	1.6	-33.3	34.7	-35.7	35.6	-38.1	36.5	-41.1	37.7		
(5)			-30.5	24.5	4.1	1.1	-33.4	25.4	-35.8	26.0	-38.2	26.7	-41.1	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-7.0	-7.3	-2.1	5.0	12.1	16.8	19.3	18.3	14.7	8.0	1.4	-4.0		
	DBStd	10.83	6.27	6.42	6.95	5.28	5.13	3.98	3.59	3.41	4.45	4.79	5.24	5.37		
	HDD10.0	2396	526	486	379	166	36	2	0	0	8	99	262	433		
	HDD18.3	4546	784	719	634	400	206	77	32	44	127	323	509	691		
	CDD10.0	1057	0	0	4	16	100	204	288	256	148	37	3	0		
	CDD18.3	163	0	0	0	0	11	29	62	42	17	2	0	0		
	CDH23.3	2041	0	0	6	12	196	372	754	485	198	19	0	0		
	CDH26.7	545	0	0	1	0	45	96	230	122	49	2	0	0		
(14) Wind	WSAvg	2.7	2.8	3.1	3.0	3.3	2.9	2.5	2.1	2.1	2.1	2.6	2.8	2.9		
(15) Precipitation	PrecAvg	852	53	39	44	76	82	101	87	78	81	84	71	50		
	PrecMax	1051	85	68	110	115	147	220	206	137	125	233	148	112		
	PrecMin	685	18	13	13	24	19	37	35	35	24	21	29	16		
	PrecStd	101	21	16	23	27	34	43	37	27	27	47	35	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	11.0	22.4	25.1	30.5	32.2	33.5	32.2	31.1	26.1	19.1	11.9		
		MCWB	7.1	7.9	14.8	14.1	19.6	22.1	22.0	21.7	20.6	17.7	12.9	10.4		
	2%	DB	5.6	5.9	15.0	22.1	27.7	30.0	31.4	30.1	27.7	22.8	14.0	7.1		
		MCWB	4.2	3.7	10.1	12.7	17.2	19.9	21.3	20.7	18.6	16.3	10.1	5.7		
	5%	DB	2.9	2.9	11.4	18.9	25.8	27.3	29.2	27.6	25.7	20.0	12.2	4.4		
		MCWB	1.9	1.3	7.2	11.1	16.4	18.4	20.5	19.4	18.1	15.4	9.4	3.5		
	10%	DB	1.4	1.2	7.9	15.2	22.8	25.4	27.3	26.2	23.3	16.6	9.1	2.5		
		MCWB	0.7	0.0	4.5	8.7	14.6	17.1	19.3	18.5	17.4	13.1	7.0	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	7.9	15.4	16.5	21.4	22.9	24.5	23.5	22.7	20.2	14.0	10.5		
		MCDB	8.2	9.6	22.1	21.3	27.8	29.7	30.1	27.8	27.7	24.4	17.4	11.4		
	2%	WB	4.3	4.0	11.4	13.9	19.7	21.4	23.1	22.3	21.1	18.2	11.4	5.6		
		MCDB	5.5	6.0	14.2	19.7	24.8	26.7	28.2	26.7	24.3	20.9	13.7	6.1		
	5%	WB	2.2	1.8	7.6	11.8	18.0	20.2	22.1	21.3	20.0	15.8	9.2	3.4		
		MCDB	2.9	3.0	10.1	17.2	23.1	24.4	26.4	25.3	23.3	18.4	11.6	4.2		
	10%	WB	0.7	0.2	4.6	9.4	16.1	19.3	21.0	20.3	18.8	13.5	7.4	1.8		
		MCDB	1.3	1.0	7.9	14.6	21.0	23.4	25.1	24.0	22.6	16.3	9.1	2.4		
(35) Mean Daily Temperature Range	MDBR		8.5	10.4	12.5	13.2	14.7	14.2	14.1	13.8	13.7	11.0	8.6	6.7		
	5% DB	MCDBR	7.8	10.8	15.6	19.2	18.0	16.6	15.9	15.8	16.1	16.0	12.7	8.4		
		MCWBR	6.8	8.7	10.4	11.0	8.9	8.1	7.4	7.8	8.5	10.1	9.3	7.3		
	5% WB	MCDBR	7.1	9.8	13.5	16.5	14.5	12.7	12.4	12.2	12.0	12.8	11.5	7.8		
		MCWBR	6.6	8.4	10.0	10.4	8.1	7.0	6.7	6.7	7.4	9.3	9.2	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.294	0.311	0.342	0.372	0.392	0.390	0.379	0.366	0.330	0.298	0.272		
	taud		2.346	2.285	2.412	2.378	2.355	2.318	2.340	2.363	2.400	2.498	2.522	2.433		
	Ebn at Noon		843	892	927	920	897	877	874	872	854	846	823	820		
	Edh at Noon		77	99	101	114	120	126	121	114	102	80	66	64		
(44) All-Sky Solar Radiation	RadAvg		1.26	2.18	3.57	4.54	5.45	5.98	6.05	5.14	4.05	2.42	1.36	0.96		
	RadStd		0.12	0.24	0.42	0.43	0.50	0.40	0.29	0.31	0.38	0.28	0.19	0.10		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page